INTELLIDOCS-AI

Resilient, Hybrid RAG Document Q&A System

THE PROBLEM

- > LLMs hallucinate when asked about private files.
- > Simple Vector Search misses exact numeric values, codes & acronyms.
- > RAG servers easily crash under free-tier memory (512MB RAM) and LLM API rate limits (HTTP 429).

KEY ACHIEVEMENT:

Reduced RAM footprint from ~800MB to ~150MB, enabling free deployment on Render while keeping Hybrid Search

THE RAG SOLUTION

- v Hybrid Retrievals: ChromaDB (HNSW Dense) + Rank-BM25 (Sparse) merged via RRF (k=60).
- v Resilient Failover Cascade: Graceful degradation from Gemini 2.0 -> 1.5 -> local mock summaries.
- v Multi-Document Compare: Unified side-by-side analysis.

PRODUCTION TECH STACK

Frontend	React 19, Vite, Tailwind CSS
Backend API	FastAPI, Uvicorn, Python 3.12
Embeddings	Google Gemini gemini-embedding-001
Vector DB	ChromaDB (Persistent HNSW)
Sparse Match	Okapi Rank-BM25
Synthesis / LLM	Google Gemini Flash Cascade
Validation	Pytest Suite (10 Passed Tests)

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Vercel: intellidocs-ai-tau.vercel.app
Render: intellidocs-api-yedx.onrender.com